

Analysis Report on Wacker Neuson Gear Pump

TSQR-0586
2026-6-29

Prepared by: Wang Kuan

Approved by:

Wacker Neuson Company:

Customer feedback failure mode: Gear pump oil leakage



I. Background Overview

Our company received feedback from your company that this gear pump exhibited an oil leakage failure. To identify the root cause, we conducted a systematic inspection and analysis of the returned unit immediately upon receipt. After preliminary visual inspection and performance retests, the results were found to differ significantly from the failure reported by the customer. Details are as follows.

II. Inspection Methods and Process

- 1. Preliminary visual inspection: Checked the pump body surface for obvious impact damage, sand holes, or scratches on seals and mating/joint faces.
Abnormal eccentric wear marks were found on the mounting pilot (spigot).
- 2. Airtightness test (core reproduction item): Performed in accordance with the standard work instruction. Observation: no abnormal condition.
- 3. Performance retest: Retested under rated pressure to measure the output flow of this gear pump.

III. Inspection Results and Data Analysis

Inspection Item	Standard / Requirement	Retest Actual Result	Conclusion
Airtightness (oil-leak reproduction)	Extended pressure hold 3 min, no leakage	No leak point found (hold pressure stable, no pressure drop)	Inconsistent with customer feedback "oil leakage"

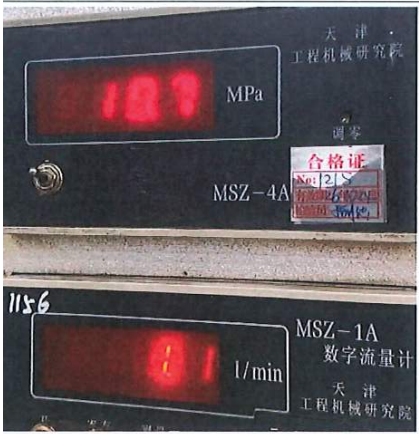
Inspection Item	Standard / Requirement	Retest Actual Result	Conclusion
Flow confirmation	19.2-21.4 L/min	Essentially no flow	Unqualified

On-site test status:

Test-bench flow confirmation:

This gear pump cannot meet the test condition: 17.2 MPa

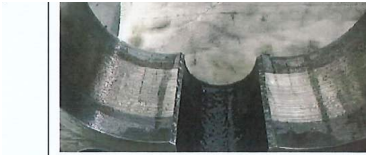
→ The factory test bench can no longer build/load pressure.



Analysis of key contradiction:

Retest results show that the flow rate is significantly lower than the rated value.

IV. Disassembly Analysis

No.	Returned Product Condition	Description
1		Nameplate information: Model: SGP1-R9647D (15CC) Production date: Oct 2025 Product No.: 04801
2		Eccentric wear marks on the mounting pilot (spigot)
3		Gear pump disassembled; secondary bore sweeping of the intermediate body → internal leakage

This indicates severe internal leakage (internal leak) wear inside the pump. However, the airtightness test did not find external leakage (external leak).

IV. Analysis of Reasons for Discrepancy Between Failure Symptoms and Test Results

Based on the contradictory phenomenon of “low flow but no external leakage” above, our analysis is as follows:

1. Differences exist between the failure operating conditions and the static test → no oil-leakage phenomenon observed.
2. Root-cause inference (key point): Severely low flow without external leakage is typical of internal pump wear. Eccentric wear marks on the mounting pilot suggest the gear pump was subjected to radial force, causing secondary internal bore sweeping, which resulted in “no flow” during the return-to-factory test.
3. The “oil leakage” reported by your company may involve “misjudgment” or other “causes.”

V. Conclusion

1. Current product status: Per our retest, the pump’s airtightness is currently qualified (no external leakage), but performance (flow) is unqualified.
2. Consistency with customer feedback: The current product status (low flow) is completely inconsistent with your company’s earlier feedback that “no abnormal operating conditions were found.” The customer’s statement of “no abnormal conditions” while claiming oil leakage is logically contradictory to the “severe internal leakage (insufficient performance)” measured by us.
3. Recommended judgment: Field misjudgment or inaccurate description of operating conditions cannot be ruled out. The fundamental failure mode of this pump is caused by internal wear, rather than simple external seal failure alone.

The above concludes this report.

Best regards,



Tianjin Shimadzu Hydraulic Co., Ltd.
Quality Assurance Department